

Circuit family $\times$ initialisation $\times$ dataset

A progressive block-size study on DIV2K, TU-Berlin and QuickDraw

Generated 2026-05-25 · mean test PSNR over the test split

Summary. We train five quantum-circuit transform families — `rich` (complex $U(4)$ gates), `qft`, `tebd`, `entangled_qft`, `mera` — under a block-size curriculum, from two initialisations (identity-operator and random), on three datasets: DIV2K (natural images, 256^2), TU-Berlin (high-res sketches, 256^2), and QuickDraw (low-res drawings, 32^2). **The dataset sets the regime:** on DIV2K the block-size curve is flat and `rich` leads; on block-sparse TU-Berlin it falls (small blocks win) and classical block-DCT is near-lossless; on coarse QuickDraw the global QFT-family wins outright and block-DCT is weak. **A common rate- crossover holds:** classical block-DCT dominates the learned circuits at light compression but they overtake it as compression tightens (and on QuickDraw the learned circuits win at every rate). `rich` leads on DIV2K, the QFT-family on the two drawing sets; **identity init is never worse than random** — often dramatically better.

1 Setup

Each *_progressive sweep trains a circuit family under a **block-size curriculum**: at stage k the inner basis is a $2^k \times 2^k$ circuit replicated across the image by a `BlockedBasis` wrapper (the bare full circuit at $k = 8$). Stages are independent — no warm-start — and each trains for 1008 steps (generalized preset, 112 epochs) under an `MSELoss` that keeps the top- k coefficients by magnitude.

Families	<code>rich</code> (complex $U(4)$ 2-qubit gates), <code>qft</code> , <code>tebd</code> , <code>entangled_qft</code> , <code>mera</code>
Initialisations	identity — every gate dropped to its manifold identity ($H \rightarrow I_2$, controlled-phase $\rightarrow$ phase 0), so the circuit starts as the identity operator. random — <code>rich</code> : Haar $U(2)/U(4)$; <code>qft</code> : Haar $U(2)$ on Hadamards + random controlled-phase angles; <code>tebd/entangled_qft/mera</code> : native seeded random gates.
Stage range	<code>rich/qft</code> : $k = 1..8$; <code>tebd/entangled_qft</code> : $k = 2..8$; <code>mera</code> : $k \in \{2, 4, 8\}$ (m must be a power of 2). Tables show $k = 2..8$.
Datasets	DIV2K-8q — 500 natural images, centre-crop + LANCZOS to 256×256 . TU-Berlin — 500 line-drawing sketches, $\approx 96.5\%$ white pixels, same preprocessing.
Metric	mean test PSNR @ keep-ratio ρ (fraction of coefficients retained); $\rho = 0.20$ is the headline rate.

2 DIV2K-8q: natural images

DIV2K is 500 natural images, centre-crop + LANCZOS to 256×256 ($m = n = 8$). Block-DCT-8 is strong here (34.0 dB @ $\rho = 0.20$).

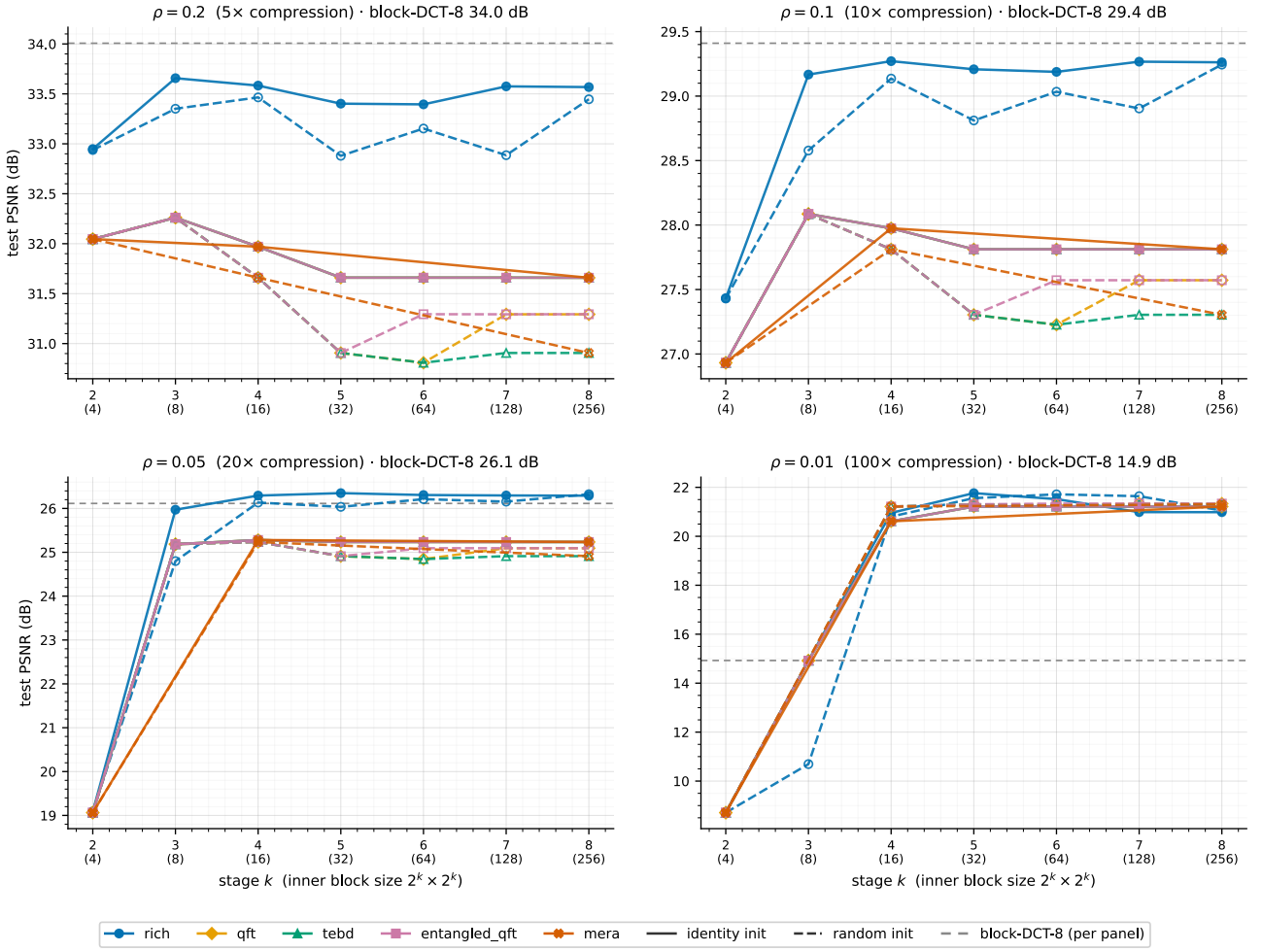


Figure 1: DIV2K: family $\times$ init PSNR vs block size k at four keep ratios. One colour per family, solid = identity / dashed = random init; grey dashed = classical block-DCT-8 (value per panel).

Findings.

- **rich leads the learned families but block-DCT wins at the light rate.** At $\rho = 0.20$ rich peaks at 33.7 dB, ≈ 1.5 dB above every other family, yet just below block-DCT-8 (34.0 dB).
- **The learned circuits overtake block-DCT as compression tightens.** block-DCT-8 falls 34.0 $\rightarrow$ 29.4 $\rightarrow$ 26.1 $\rightarrow$ 14.9 dB across $\rho = 0.20/0.10/0.05/0.01$; rich draws level by $\rho = 0.05$ and every family clears it by ≈ 6 –7 dB at $\rho = 0.01$.
- **Under identity init the four QFT-derived families are bit-identical** at every k (gate counts at $k = 8$: qft 72, tebd 32, entangled_qft 80, mera 44) – they all converge to the same QFT operator.
- **Identity init $\geq$ random**; the curve is flat in k (≈ 33 dB rich, ≈ 31.7 rest) at $\rho = 0.20$.

Per-rate tables (mean test PSNR, dB). $\rho = 0.20 - 5\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	32.9	33.7	33.6	33.4	33.4	33.6	33.6
rich	random	32.9	33.4	33.5	32.9	33.2	32.9	33.4
qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
qft	random	32	32.3	31.7	30.9	30.8	31.3	31.3
tebd	identity	32	32.3	32	31.7	31.7	31.7	31.7
tebd	random	32	32.3	31.7	30.9	30.8	30.9	30.9
ent-qft	identity	32	32.3	32	31.7	31.7	31.7	31.7
ent-qft	random	32	32.3	31.7	30.9	31.3	31.3	31.3
mera	identity	32	n/a	32	n/a	n/a	n/a	31.7
mera	random	32	n/a	31.7	n/a	n/a	n/a	30.9

$\rho = 0.10 - 10\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	27.4	29.2	29.3	29.2	29.2	29.3	29.3
rich	random	27.4	28.6	29.1	28.8	29	28.9	29.2
qft	identity	26.9	28.1	28	27.8	27.8	27.8	27.8
qft	random	26.9	28.1	27.8	27.3	27.2	27.6	27.6
tebd	identity	26.9	28.1	28	27.8	27.8	27.8	27.8
tebd	random	26.9	28.1	27.8	27.3	27.2	27.3	27.3
ent-qft	identity	26.9	28.1	28	27.8	27.8	27.8	27.8
ent-qft	random	26.9	28.1	27.8	27.3	27.6	27.6	27.6
mera	identity	26.9	n/a	28	n/a	n/a	n/a	27.8
mera	random	26.9	n/a	27.8	n/a	n/a	n/a	27.3

$\rho = 0.05 - 20\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	19.1	26	26.3	26.3	26.3	26.3	26.3
rich	random	19.1	24.8	26.1	26	26.2	26.2	26.3
qft	identity	19.1	25.2	25.3	25.2	25.2	25.2	25.2
qft	random	19.1	25.2	25.2	24.9	24.8	25.1	25.1
tebd	identity	19.1	25.2	25.3	25.2	25.2	25.2	25.2
tebd	random	19.1	25.2	25.2	24.9	24.8	24.9	24.9
ent-qft	identity	19.1	25.2	25.3	25.2	25.2	25.2	25.2
ent-qft	random	19.1	25.2	25.2	24.9	25.1	25.1	25.1
mera	identity	19.1	n/a	25.3	n/a	n/a	n/a	25.2
mera	random	19.1	n/a	25.2	n/a	n/a	n/a	24.9

$\rho = 0.01 - 100\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	8.7	14.9	21	21.8	21.5	21	21
rich	random	8.7	10.7	20.8	21.6	21.7	21.6	21.1
qft	identity	8.7	14.9	20.6	21.2	21.2	21.2	21.2
qft	random	8.7	14.9	21.2	21.3	21.3	21.3	21.3
tebd	identity	8.7	14.9	20.6	21.2	21.2	21.2	21.2
tebd	random	8.7	14.9	21.2	21.3	21.3	21.3	21.3
ent-qft	identity	8.7	14.9	20.6	21.2	21.2	21.2	21.2
ent-qft	random	8.7	14.9	21.2	21.3	21.3	21.3	21.3
mera	identity	8.7	n/a	20.6	n/a	n/a	n/a	21.2
mera	random	8.7	n/a	21.2	n/a	n/a	n/a	21.3

3 TU-Berlin: sketches (a different regime)

Sketches are **block-sparse** — most 8×8 tiles are blank — so a classical block transform is near-lossless: block-DCT-8 reaches 100.7 dB mean / 94.5 median at $\rho = 0.20$ (versus ≈ 32 on DIV2K), but collapses to ≈ 4.8 dB at $\rho = 0.01$.

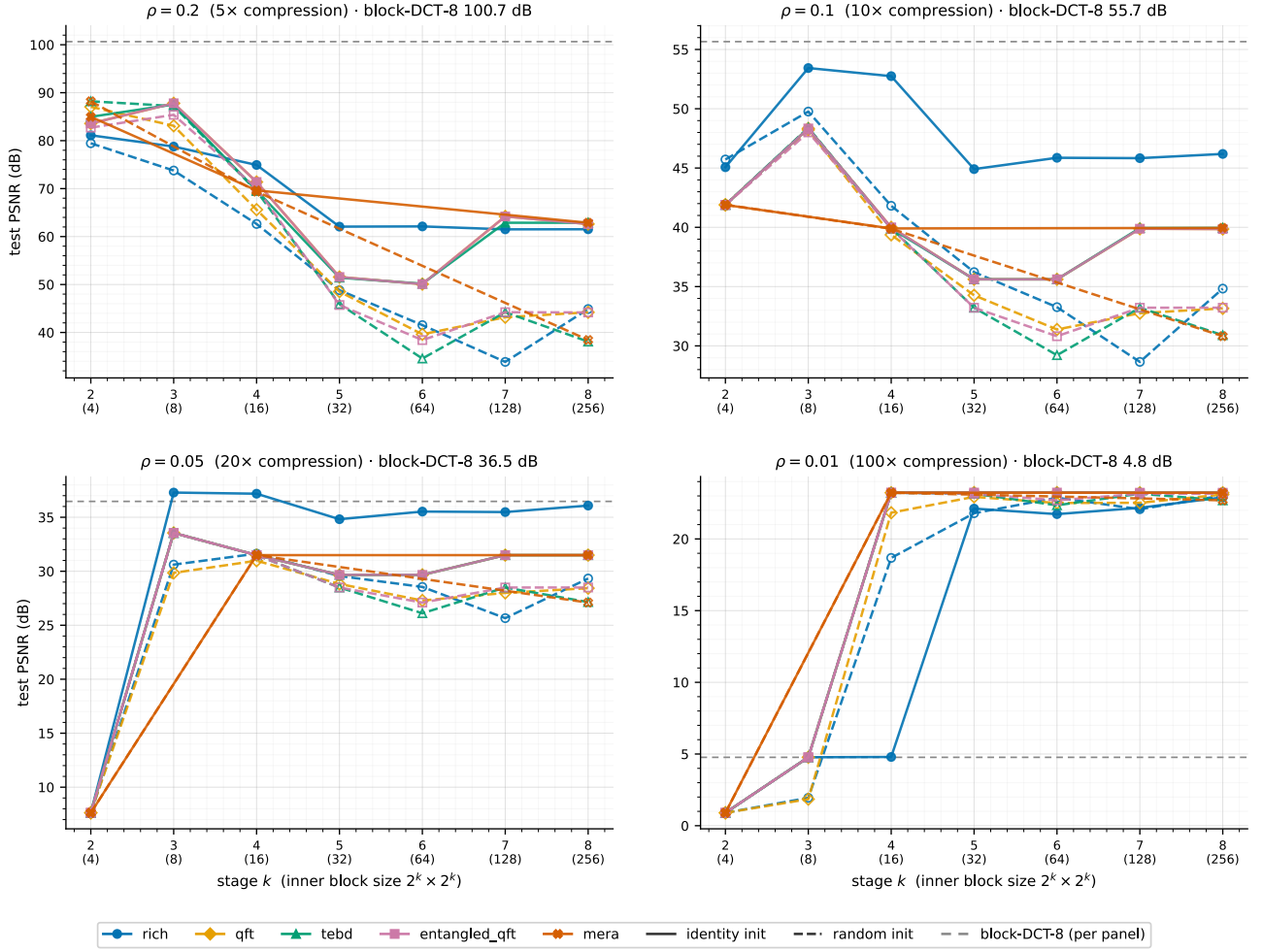


Figure 2: TU-Berlin: family $\times$ init PSNR vs block size k at four keep ratios. Note the **falling** curve (small blocks win) and the very high absolute PSNR; grey dashed = block-DCT-8 (value per panel).

Findings.

- **The block-size curve inverts at the light rate.** Small blocks win — each blank tile costs almost no coefficients — the opposite of DIV2K’s flat curve.
- **The family ranking flips with the rate.** At $\rho = 0.20$ the QFT-derived families peak ≈ 88 dB at $k = 3$, above rich (≈ 79); at $\rho = 0.05$ rich (identity) leads (≈ 37 dB); at $\rho = 0.01$ the QFT-derived families edge back ahead (≈ 23.2 vs rich 22.9).
- **The heavy-compression crossover is extreme here.** block-DCT-8 falls $100.7 \rightarrow 55.7 \rightarrow 36.5 \rightarrow 4.8$ dB across $\rho = 0.20/0.10/0.05/0.01$: near-lossless at the light rate, but the learned circuits beat it by ≈ 18 dB at $\rho = 0.01$.
- **Identity $\triangleright$ random** — the gap reaches ≈ 25 – 30 dB at large k , far wider than on DIV2K. qft $\equiv$ entangled_qft at all k under identity init.

Per-rate tables (mean test PSNR, dB). $\rho = 0.20 - 5\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	81.1	78.8	74.9	62.1	62.1	61.5	61.5
rich	random	79.5	73.8	62.6	48.8	41.5	33.9	44.9
qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
qft	random	87.1	83.1	65.6	48.5	39.6	43.2	44.2
tebd	identity	85	87.5	69.6	51.4	50.2	62.9	62.9
tebd	random	88.2	87.2	69.5	45.9	34.6	44.2	38.1

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
ent-qft	identity	83.6	87.8	71.4	51.6	50.1	64.2	62.6
ent-qft	random	82.7	85.4	70.1	45.7	38.4	44.2	44.2
mera	identity	85	n/a	69.6	n/a	n/a	n/a	62.9
mera	random	88.2	n/a	69.4	n/a	n/a	n/a	38.4

$\rho = 0.10 - 10\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	45.1	53.4	52.7	44.9	45.9	45.8	46.2
rich	random	45.7	49.8	41.8	36.2	33.3	28.6	34.8
qft	identity	41.9	48.3	40	35.6	35.6	39.9	39.9
qft	random	41.9	48.2	39.4	34.3	31.4	32.8	33.2
tebd	identity	41.9	48.4	39.9	35.6	35.6	39.9	39.9
tebd	random	41.9	48.4	39.9	33.2	29.2	33.2	30.9
ent-qft	identity	41.9	48.3	40	35.6	35.6	39.9	39.9
ent-qft	random	41.9	48	39.9	33.2	30.8	33.2	33.2
mera	identity	41.9	n/a	39.9	n/a	n/a	n/a	39.9
mera	random	41.9	n/a	39.9	n/a	n/a	n/a	30.8

$\rho = 0.05 - 20\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	7.6	37.3	37.2	34.8	35.5	35.5	36.1
rich	random	7.6	30.6	31.6	29.6	28.5	25.7	29.3
qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
qft	random	7.6	29.8	31	28.8	27.3	28	28.4
tebd	identity	7.6	33.5	31.5	29.7	29.7	31.5	31.5
tebd	random	7.6	33.5	31.5	28.5	26.1	28.5	27.1
ent-qft	identity	7.6	33.6	31.5	29.7	29.7	31.5	31.5
ent-qft	random	7.6	33.5	31.5	28.5	27.1	28.5	28.5
mera	identity	7.6	n/a	31.5	n/a	n/a	n/a	31.5
mera	random	7.6	n/a	31.5	n/a	n/a	n/a	27.1

$\rho = 0.01 - 100\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)	$k=6$ (64)	$k=7$ (128)	$k=8$ (256)
rich	identity	0.9	4.8	4.8	22.1	21.7	22.2	22.9
rich	random	0.9	1.9	18.7	21.8	22.9	22.1	23
qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
qft	random	0.9	1.9	21.8	22.9	22.5	22.5	23.1
tebd	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
tebd	random	0.9	4.8	23.2	23.2	22.3	23.2	22.7
ent-qft	identity	0.9	4.8	23.2	23.2	23.2	23.2	23.2
ent-qft	random	0.9	4.8	23.2	23.2	22.7	23.2	23.2
mera	identity	0.9	n/a	23.2	n/a	n/a	n/a	23.2
mera	random	0.9	n/a	23.2	n/a	n/a	n/a	22.7

4 QuickDraw: low-resolution drawings

QuickDraw is 32×32 ($m = n = 5$), so the curriculum spans only $k = 1..5$ (block sizes 2..32; tables show $k = 2..5$, mera at $k \in \{2, 4\}$). The drawings are sparse strokes on a **dark** background, but at this low resolution they are far less block-

compressible than the high-res TU-Berlin sketches: classical block-DCT-8 reaches only 26.63 dB at $\rho = 0.20$ (vs ≈ 100 on TU-Berlin), so the learned circuits beat it.

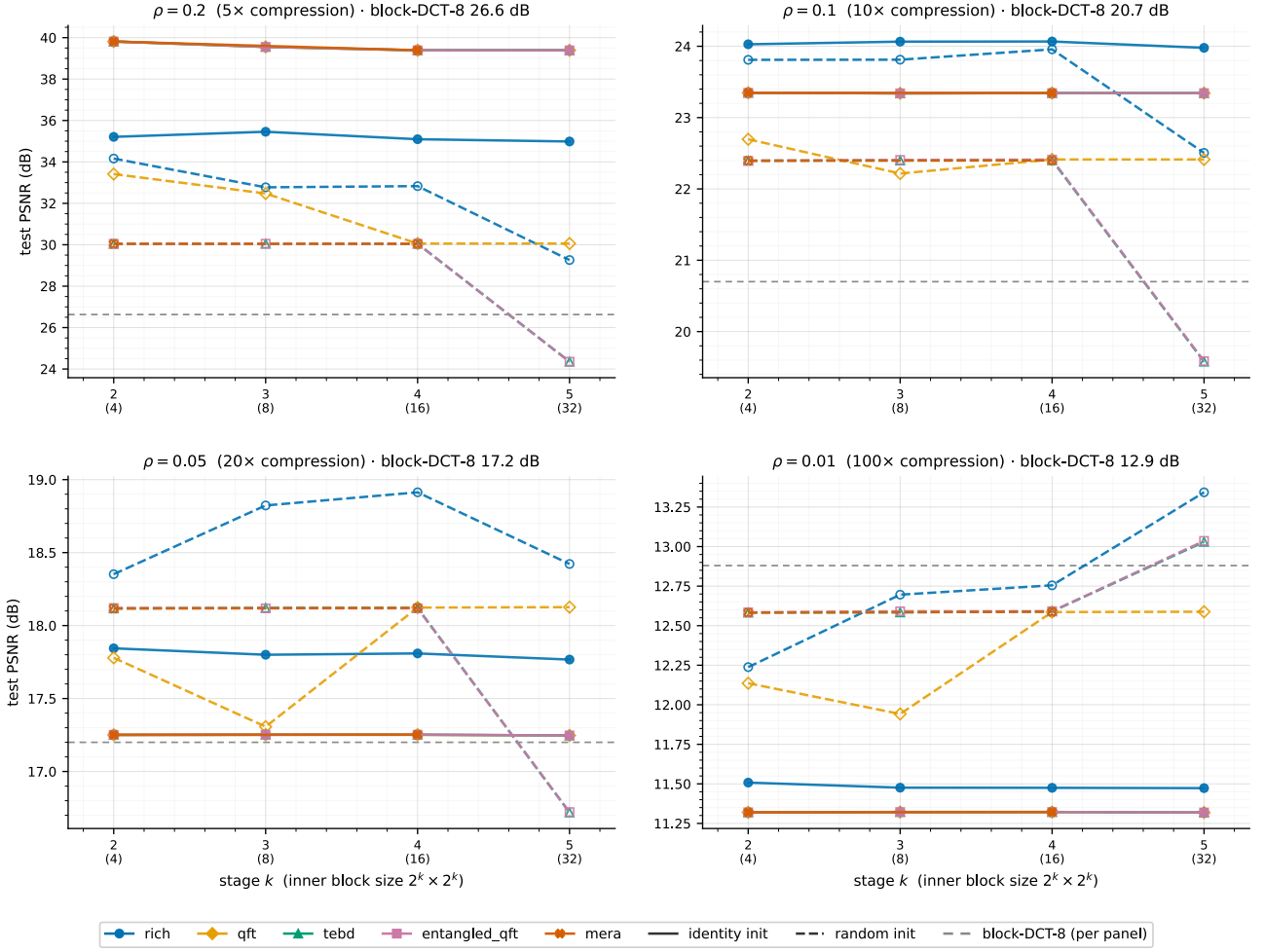


Figure 3: QuickDraw: family $\times$ init PSNR vs block size k at four keep ratios. The QFT-derived families (identity) dominate and the curve is flat in k ; grey dashed = block-DCT-8 (value per panel), which the learned circuits clear at every rate.

Findings.

- **Learned circuits beat block-DCT at every rate.** Unlike the 256² sets, block-DCT-8 is weak on 32 $\times$ 32 drawings (26.6 dB @ $\rho = 0.20$), so the QFT-derived families (≈ 39.5) and rich (≈ 35) both clear it.
- **The QFT-derived families lead and are bit-identical under identity init**, beating rich by ≈ 4 dB at $\rho = 0.20$. The curve is flat in k (like DIV2K, unlike TU-Berlin’s falling curve).
- **At heavy compression the families converge** to block-DCT (≈ 17 dB at $\rho = 0.05$, ≈ 12 at $\rho = 0.01$) and the identity advantage erodes — random init even edges ahead at some k .

Per-rate tables (mean test PSNR, dB). $\rho = 0.20 - 5\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)
rich	identity	35.2	35.5	35.1	35
rich	random	34.2	32.8	32.8	29.3
qft	identity	39.8	39.6	39.4	39.4
qft	random	33.4	32.5	30.1	30.1
tebd	identity	39.8	39.6	39.4	39.4
tebd	random	30	30	30	24.4
ent-qft	identity	39.8	39.6	39.4	39.4
ent-qft	random	30	30.1	30.1	24.4
mera	identity	39.8	n/a	39.4	n/a
mera	random	30	n/a	30	n/a

$\rho = 0.10 - 10\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)
rich	identity	24	24.1	24.1	24
rich	random	23.8	23.8	24	22.5
qft	identity	23.3	23.3	23.3	23.3
qft	random	22.7	22.2	22.4	22.4
tebd	identity	23.3	23.3	23.3	23.3
tebd	random	22.4	22.4	22.4	19.6
ent-qft	identity	23.3	23.3	23.3	23.3
ent-qft	random	22.4	22.4	22.4	19.6
mera	identity	23.3	n/a	23.3	n/a
mera	random	22.4	n/a	22.4	n/a

$\rho = 0.05 - 20\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)
rich	identity	17.8	17.8	17.8	17.8
rich	random	18.4	18.8	18.9	18.4
qft	identity	17.3	17.3	17.3	17.2
qft	random	17.8	17.3	18.1	18.1
tebd	identity	17.3	17.3	17.3	17.2
tebd	random	18.1	18.1	18.1	16.7
ent-qft	identity	17.3	17.3	17.3	17.2
ent-qft	random	18.1	18.1	18.1	16.7
mera	identity	17.3	n/a	17.3	n/a
mera	random	18.1	n/a	18.1	n/a

$\rho = 0.01 - 100\times$

family	init	$k=2$ (4)	$k=3$ (8)	$k=4$ (16)	$k=5$ (32)
rich	identity	11.5	11.5	11.5	11.5
rich	random	12.2	12.7	12.8	13.3
qft	identity	11.3	11.3	11.3	11.3
qft	random	12.1	11.9	12.6	12.6
tebd	identity	11.3	11.3	11.3	11.3
tebd	random	12.6	12.6	12.6	13
ent-qft	identity	11.3	11.3	11.3	11.3
ent-qft	random	12.6	12.6	12.6	13
mera	identity	11.3	n/a	11.3	n/a
mera	random	12.6	n/a	12.6	n/a

5 Cross-dataset discussion

	DIV2K (natural, 256 ²)	TU-Berlin (sketch, 256 ²)	QuickDraw (drawing, 32 ²)
block-size curve	flat in k	falls with k (small blocks win)	flat in k
best learned family	rich	QFT-family @ 0.2 & 0.01; rich @ 0.05	QFT-family
block-DCT @ $\rho=0.2$	34 dB (beats all)	94 dB (beats all)	27 dB (learned beat it)
vs block-DCT	learned overtake by $\rho=0.01$	learned overtake @ 0.05 & 0.01	learned win at all ρ @ 0.2; converge @ 0.01

identity vs random	identity $\geq$ random (< 1 dB)	identity $\triangleright$ random (25–30 dB)	identity wins @ 0.2; ties @ 0.05/0.01
PSNR @ $\rho=0.2$	$\approx 31\text{--}34$	$\approx 60\text{--}88$	$\approx 35\text{--}40$

The contrast is governed by **energy compaction relative to the data’s sparsity structure and resolution**. Natural-image energy is spread across scales, so a large circuit and the complex- $U(4)$ richness of rich help; high-res sketches are block-sparse, so a small per-block transform is near-lossless and the curve falls with k ; low-res QuickDraw drawings are too coarse for 8×8 blocks to exploit, so the global QFT-family solution wins outright and block-DCT is weak. A common thread: the fixed block transform degrades faster than the learned circuits as the rate tightens, so learning pays off most at aggressive compression (and, on coarse QuickDraw, at every rate).

6 Conclusions

1. **The dataset sets the regime.** “Which circuit family is best” does not transfer across natural images, high-res sketches and low-res drawings; even the sign of the block-size trend and the leading family change.
2. **Learned circuits beat classical block-DCT at aggressive compression** (and at every rate on coarse QuickDraw, where block-DCT is weak); on the 256^2 sets block-DCT wins at the light rate but the learned transforms overtake it as ρ shrinks.
3. **Architectural richness is dataset-dependent:** complex $U(4)$ (rich) leads on natural images, but the plain QFT-family solution leads on both drawing sets.
4. **Identity initialisation dominates random** on the 256^2 sets (decisively on sketches); on coarse QuickDraw the advantage holds only at the light rate and ties under heavy compression. The structured identity start is a strong, cheap prior.

Data: `results/family_init_matrix/{div2k_8q,tuberlin_8q,quickdraw_5q}/<family>_<init>/. Reproduce a sweep with experiments/qft_progressive.py --dataset <d> --family <f> --init <i>`; regenerate figures with each report’s `render_fig*.py`.